CS & IT ENGINEERING

Digital Logic

Combinational Circuits

Lecture No. 03

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Recap of Previous Lecture

Topic

n-bit Comparators

Generalised

Topics to be Covered

Topic

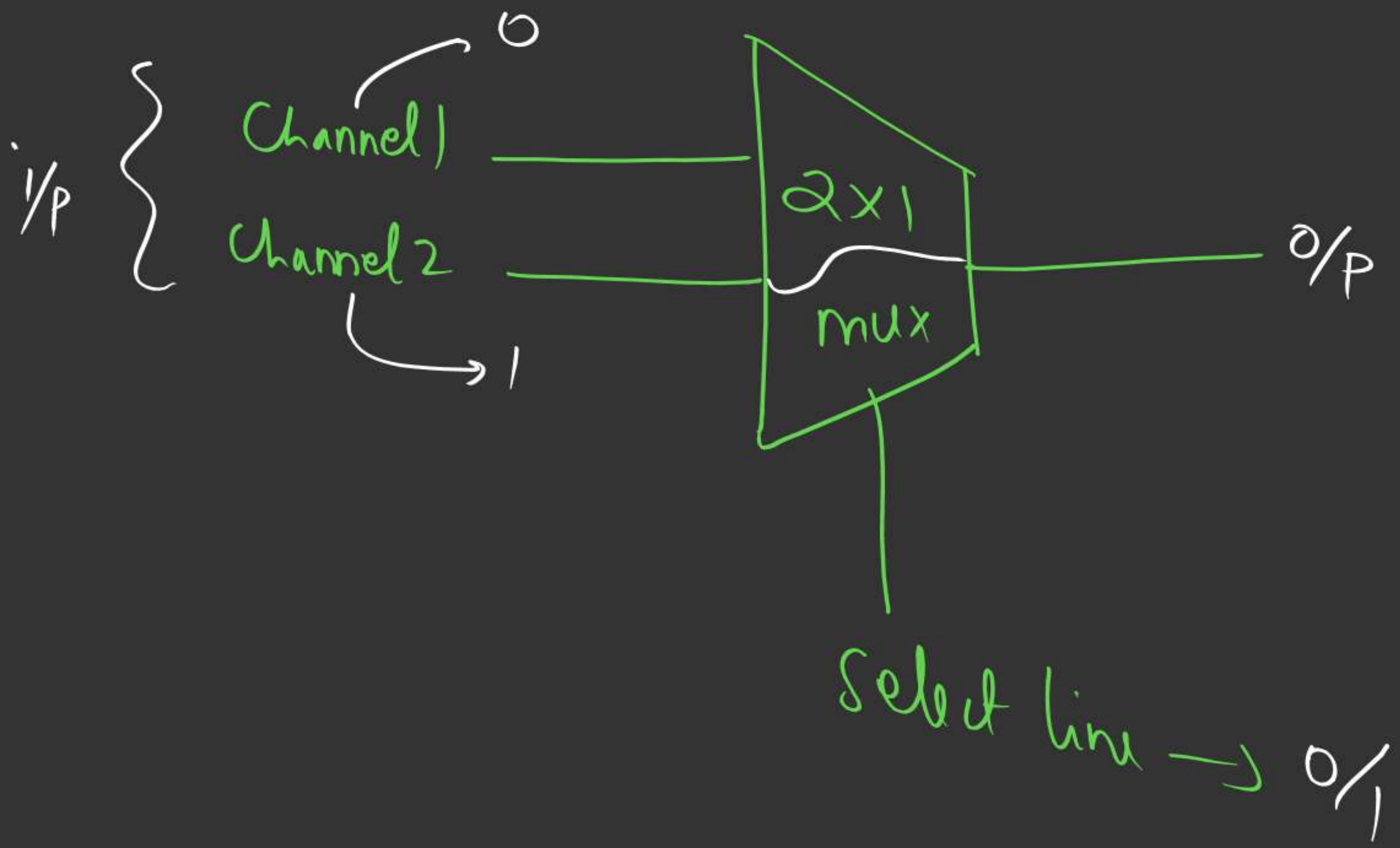
Multiplexors

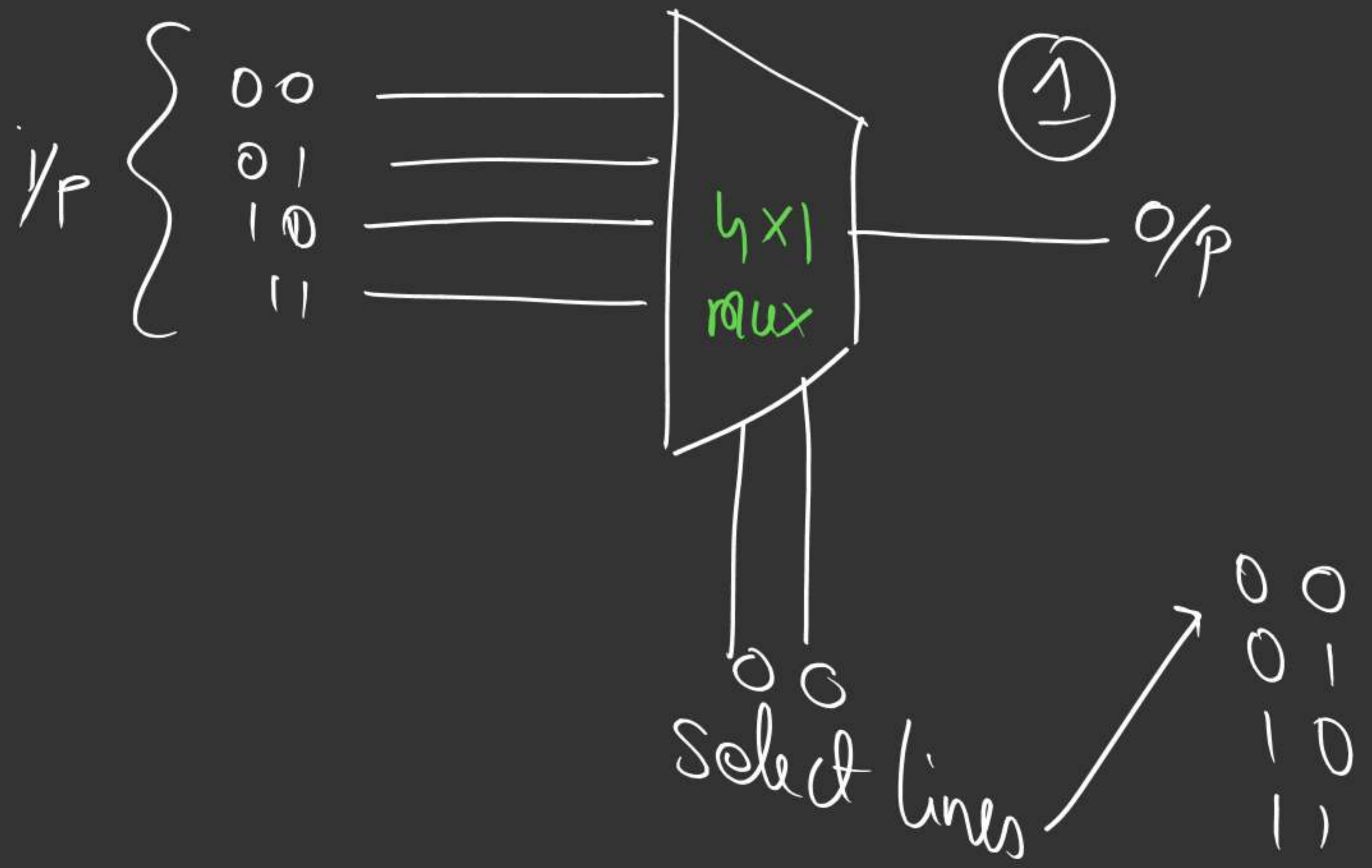
(Mux)

Multiplexer (MUX)

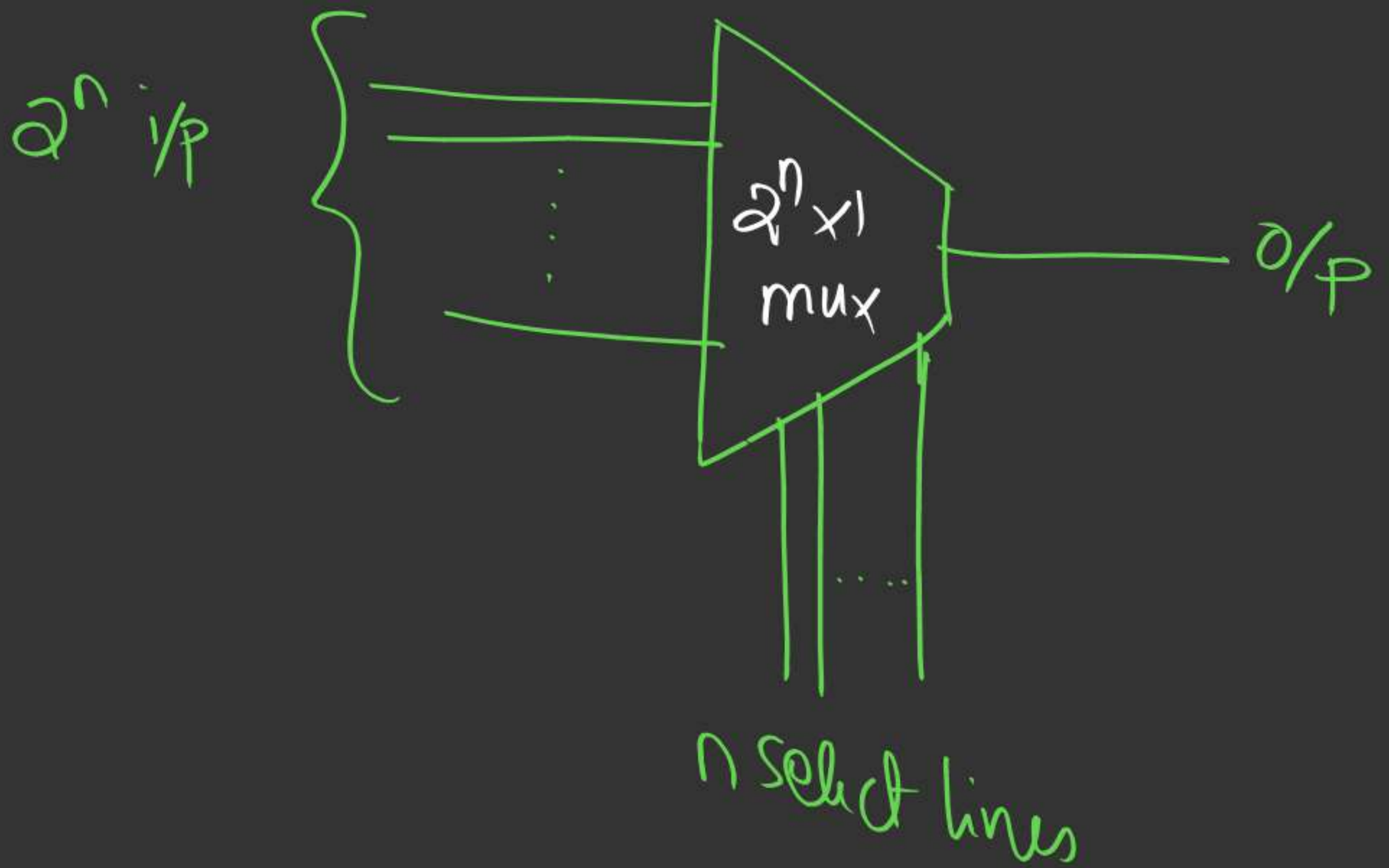
↳ multi i/p $\xrightarrow{\text{to}}$ one o/p

Select lines


0
i/p
Channel 1
Channel 2
1
2x1
mux
o/p
Select line → 0/1


I/P
00
01
10
11
4x1
mux
1
O/P
00
Select lines
00
01
10
11

MUX (in general)



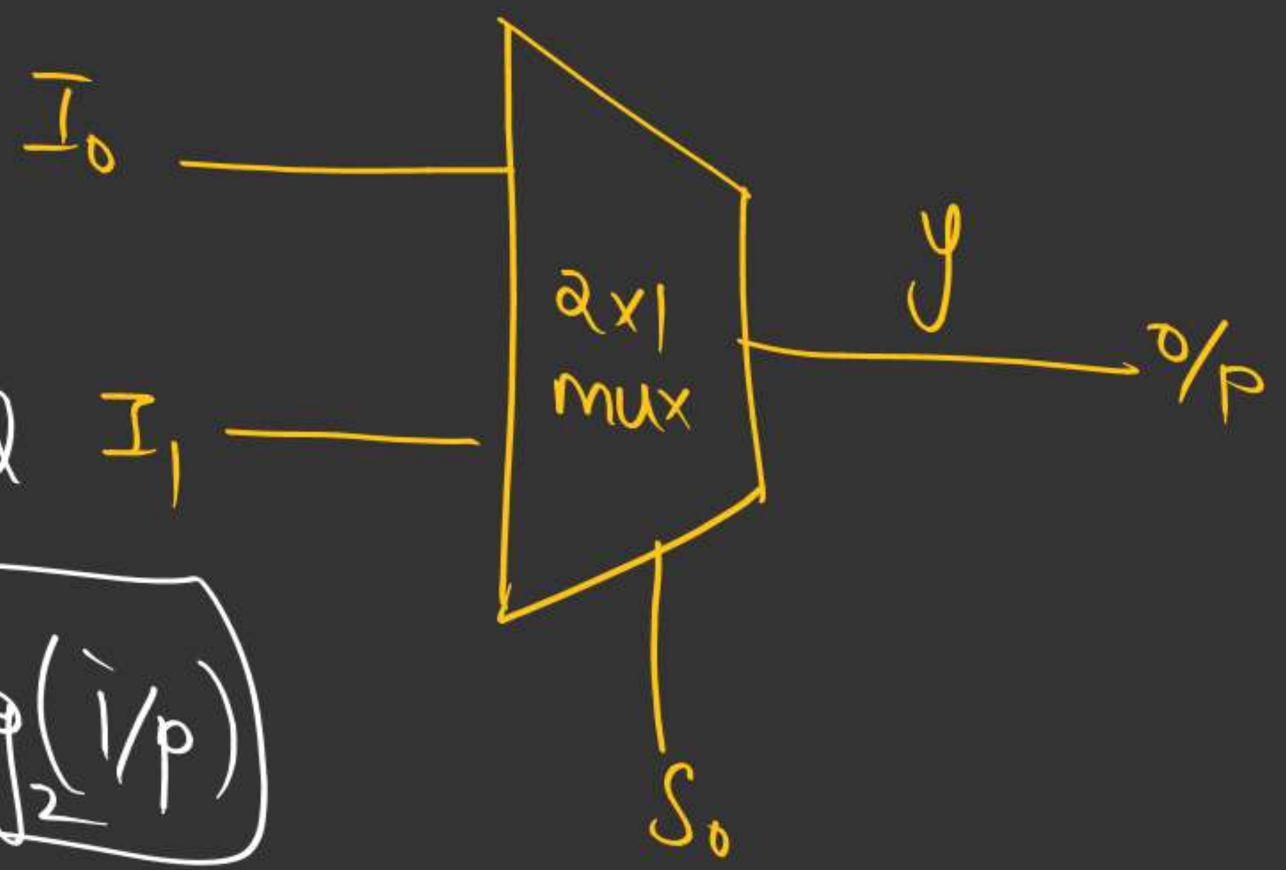
$2^n \times 1$ mux

→ 2^n to 1 converter.

→ Universal logic

→ $\boxed{\text{AND - OR}}$

1) 2x1 mux



Step 1: 2 i/p 1 o/p

2 i/p → 1 select line

In general

$$\boxed{SL = \log_2(i/p)}$$

Step 2 :- Truth Table :-

S_0	y
0	I_0
1	I_1

Step 3 :- Logical Expression:

$S_0 = 0 \longrightarrow I_0$ (O/P)

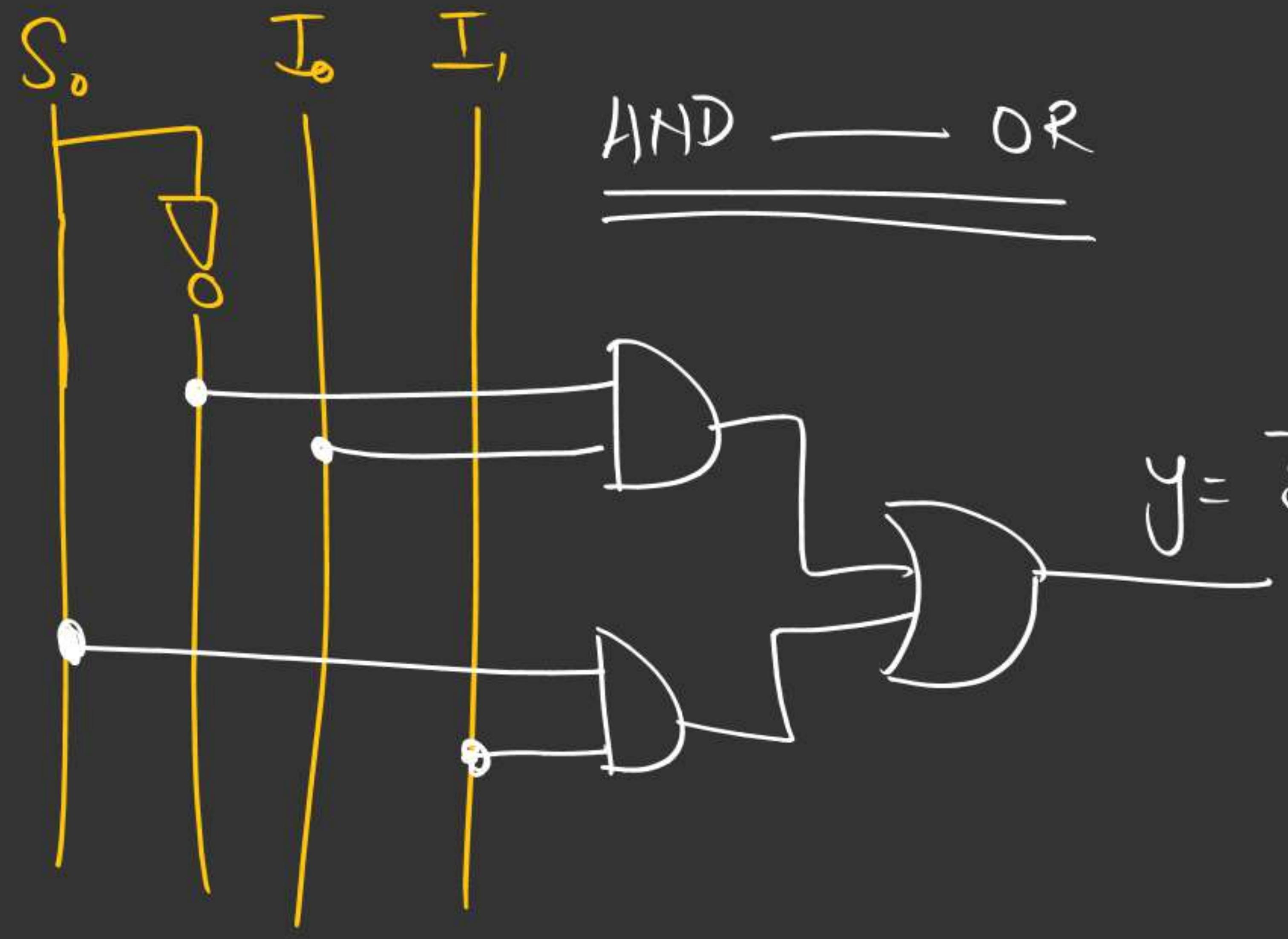
$S_0 = 1 \longrightarrow I_1$

$$\boxed{y = \overline{S_0} \cdot I_0 + S_0 \cdot I_1}$$

Step 4 :- Minimize Expression

$$\boxed{y = \bar{S_0} \cdot I_0 + S_0 \cdot I_1}$$

Step 5:- Draw logical Diagram/Hardware.

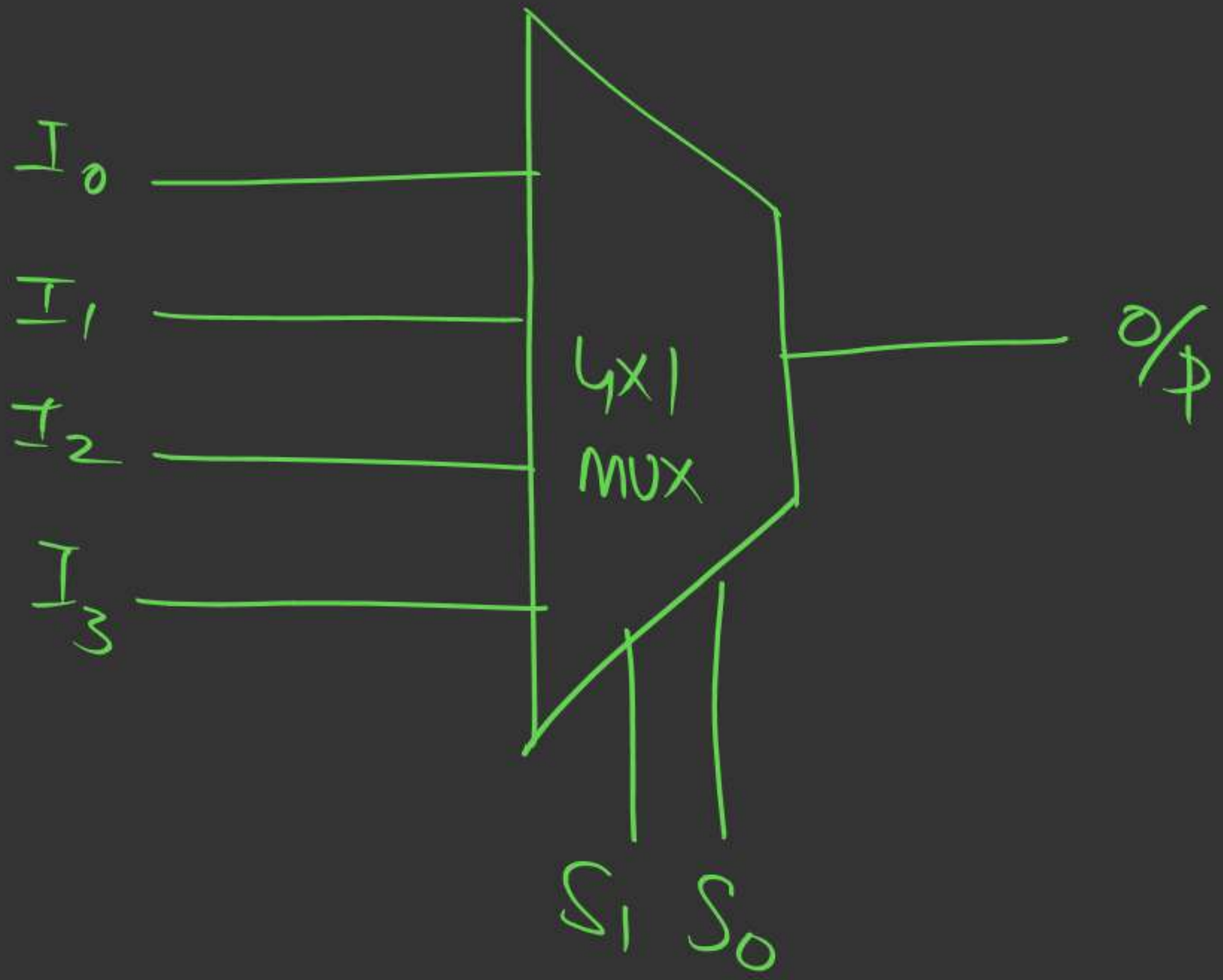


$$y = \bar{S}_0 I_0 + S_0 I_1$$

② 4×1 mux

Step 1 : 4 i/p (I_0, I_1, I_2, I_3), 1 o/p

Select lines = 2



S_2 S_1 S_0

$\overrightarrow{S_2 \, S_1 \, S_0}$

Step2:- Truth table

S_1	S_0	y
0	0	I_0
0	1	I_1
1	0	I_2
1	1	I_3

Step 3:- Logical Expression:-

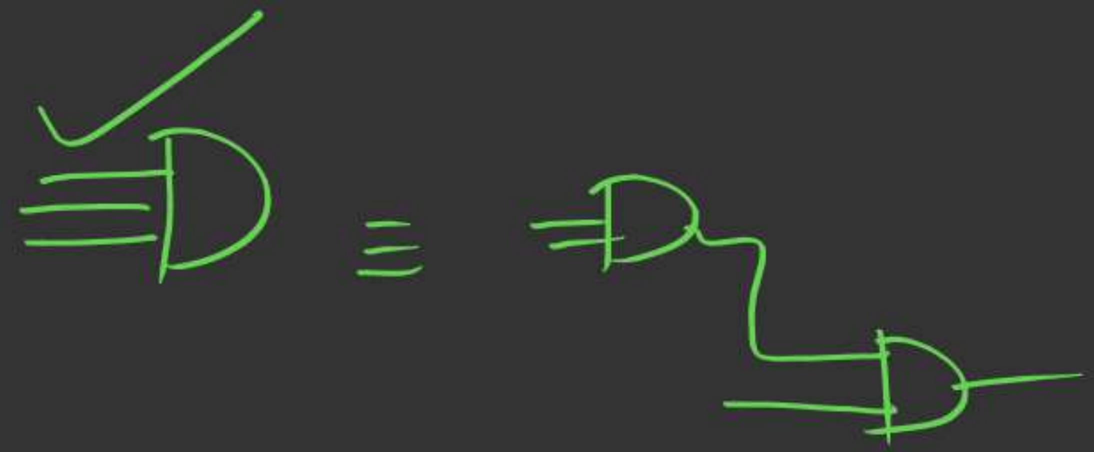
$$y = \bar{S}_1 \bar{S}_0 I_0 + \bar{S}_1 S_0 I_1 + S_1 \bar{S}_0 I_2 + S_1 S_0 I_3$$

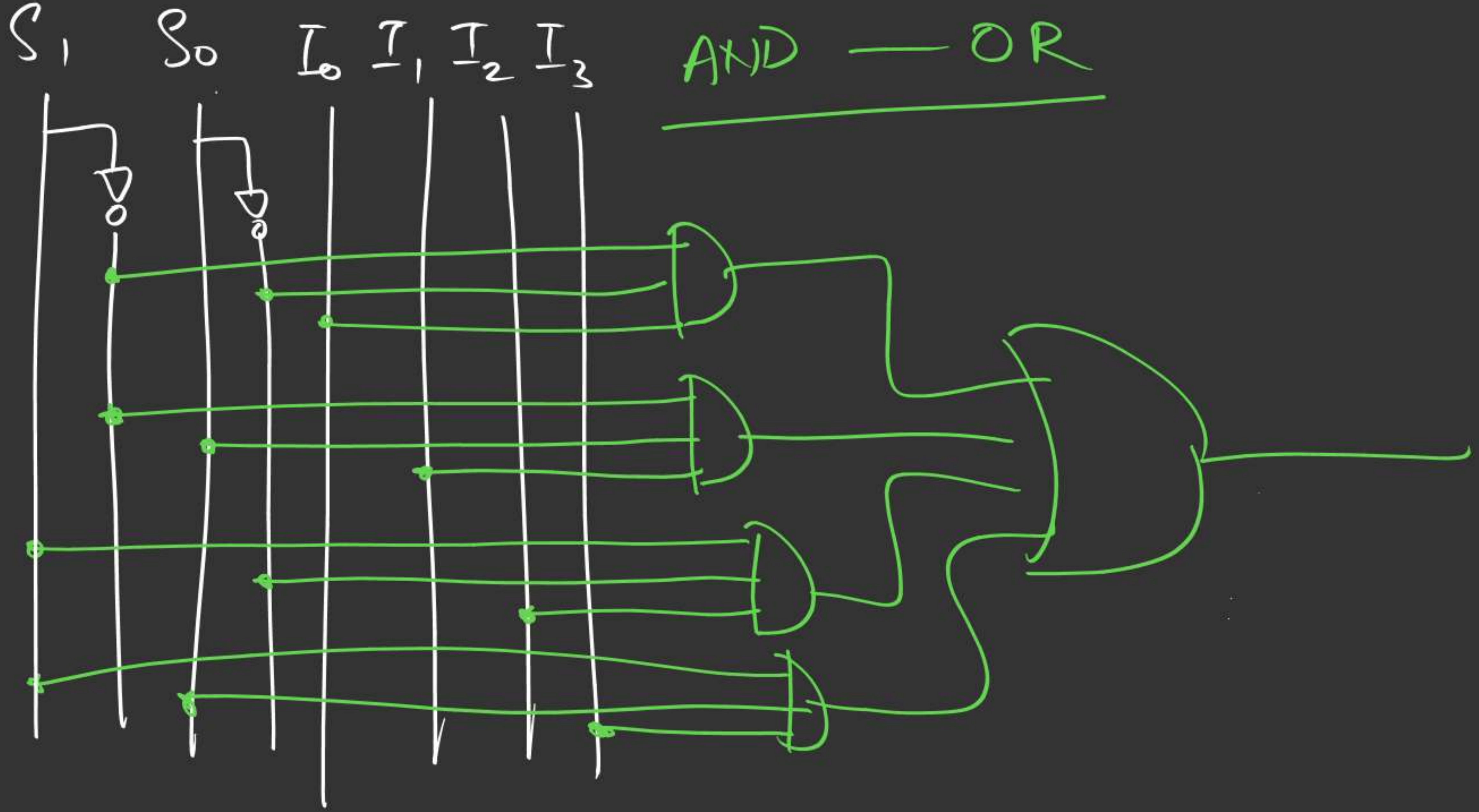
0 0
0 1
1 0
1 1

Step 4: Minimized Expression

$$Y = \overline{S_1}\,\overline{S_0}\,I_0 + \overline{S_1}\,S_0\,I_1 + S_1\,\overline{S_0}\,I_2 + S_1\,S_0\,I_3$$

Step5 :- Hardware/Diagram





AND — OR

H.W

1) Design 8x1 MUX (5 steps)

2) " 16x1 MUX (5 steps)

Type 1

Create/design a Higher order mux

by using smaller order mux

2x1 mux → Design 4x1 mux

(Q.1) How many 2x1 mux needed to design a 4x1 mux?

(Q.2) Design Structure?

eg:-

Design a $\underline{\underline{4\times1}}$ mux using 2×1 mux(s)

i/p

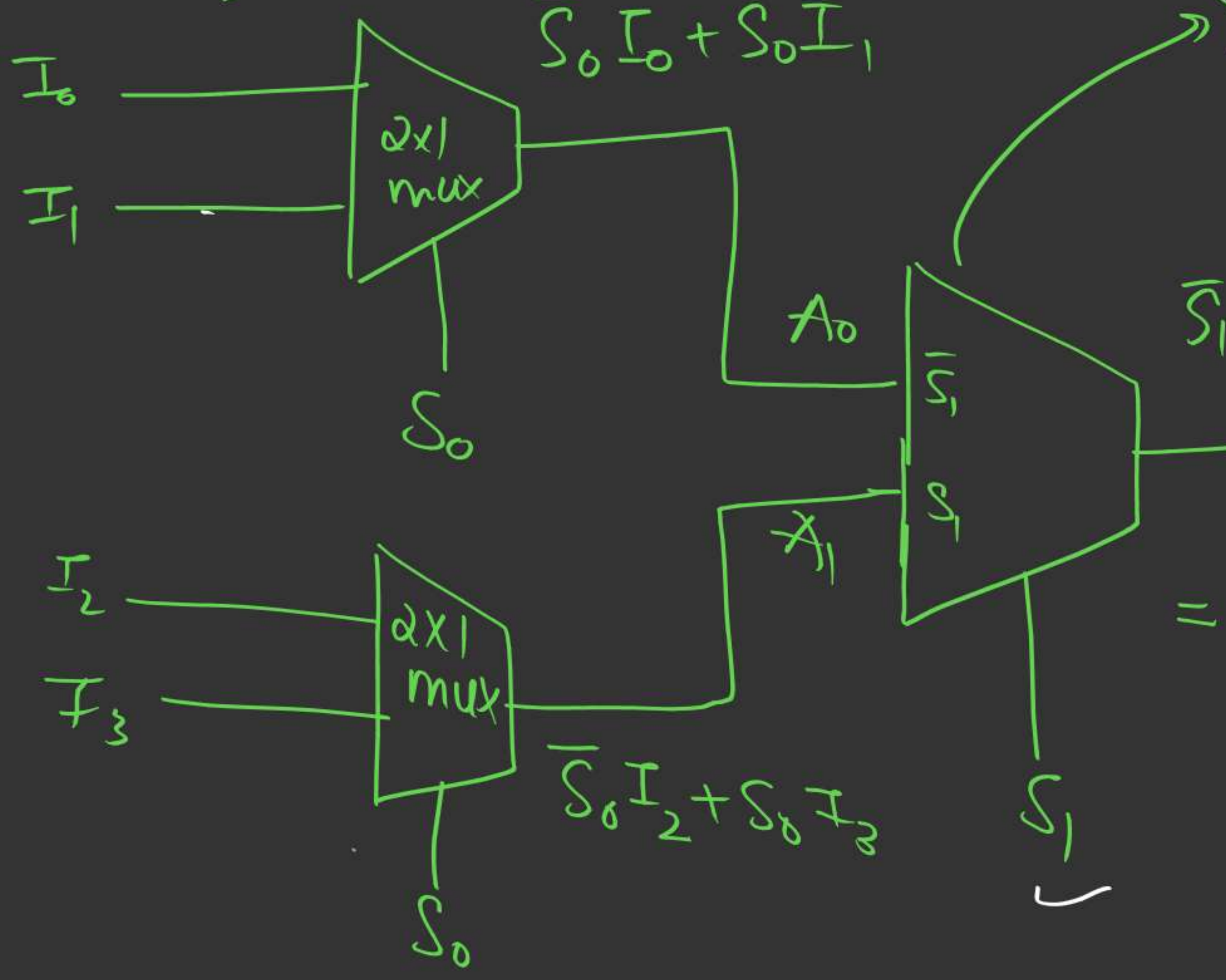
4 i/p

2 i/ps

(Q) How many 2×1 mux $\longrightarrow$ 4×1 mux.

explanation

$$\left(\bar{S}_1 \cdot A_0 + S_1 A_1\right)$$



$$\bar{S}_1\left(\bar{S}_0 I_0 + S_0 I_1\right) + S_1\left(\bar{S}_0 I_2 + S_0 I_3\right)$$

$$= \left[\bar{S}_1 \bar{S}_0 I_0 + \bar{S}_1 S_0 I_1 + S_1 \bar{S}_0 I_2 + S_1 S_0 I_3\right]$$

3 2x1 mux needed

Ans:- Shortcut (to solve this)

4×1

2×1

$$\frac{4}{2} = 2$$

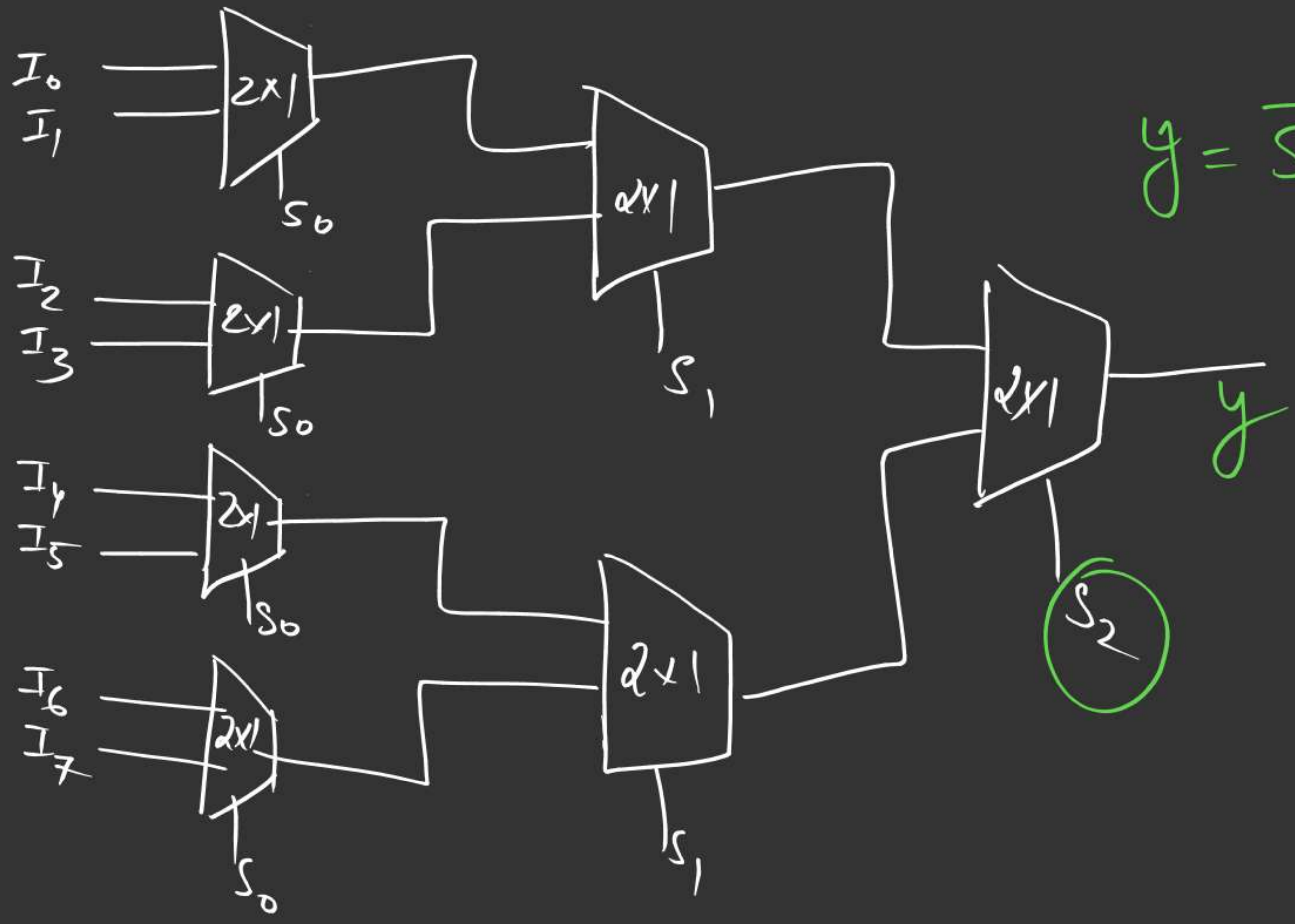
$$\frac{2}{2} = 1$$

keep dividing by smaller mux size until reach ≤ 1.

$$\text{Total} = \frac{4}{2} + \frac{2}{2} = 2 + 1 = 3 \quad (2 \times 1 \text{ mux})$$

(Q.2) How many 2×1 mux $\longrightarrow$ 8×1 mux ?

↳ Outline of Design



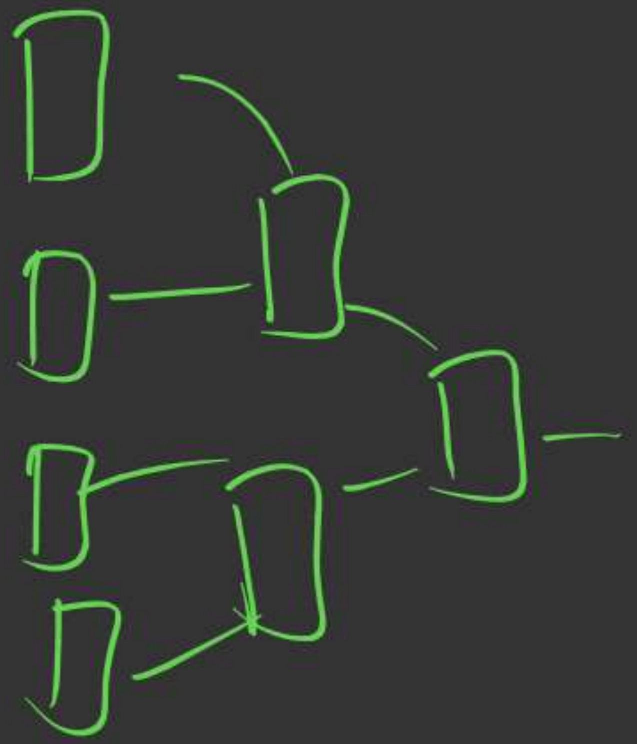
$y = \bar{S}_2(\quad) + S_2(\quad)$

Ans 7

Shortcut

8x1

2x1



$$\frac{8}{2} = 4$$

$$\frac{4}{2} = 2$$

$$\frac{2}{2} = 1$$

Total 2x1 mux = 4+2+1

= 7

Apply Shortcut

1> How many $2 \times 1 \longrightarrow 16 \times 1$?

2> " $2 \times 1 \longrightarrow 64 \times 1$?

Soln 1) 16X1 using 2X1

$\frac{16}{2} = 8$ ✓

$\frac{8}{2} = 4$ ✓

$4/2 = 2$ ✓

$2/2 = 1$ ✓

$\log_2(16)$

$= \underline{4 \text{ SL}}$ ✓

Total $= 8+4+2+1$

$= 15$

Soln2:-

64x1 using 2x1

$$\frac{64}{2} = 32$$

$$\frac{32}{2} = 16$$

$$\frac{16}{2} = 8$$

$$8/2 = 4$$

$$4/2 = 2$$

$$2/2 = 1$$

$$\text{Total} = 32 + 16 + 8 + 4 + 2 + 1$$

$$= \underline{\underline{63}}$$

Property :

How may 2x1 mux needed to design a $2^n \times 1$ mux ?

$4 \times 1 \longrightarrow 3$

$8 \times 1 \longrightarrow 7$

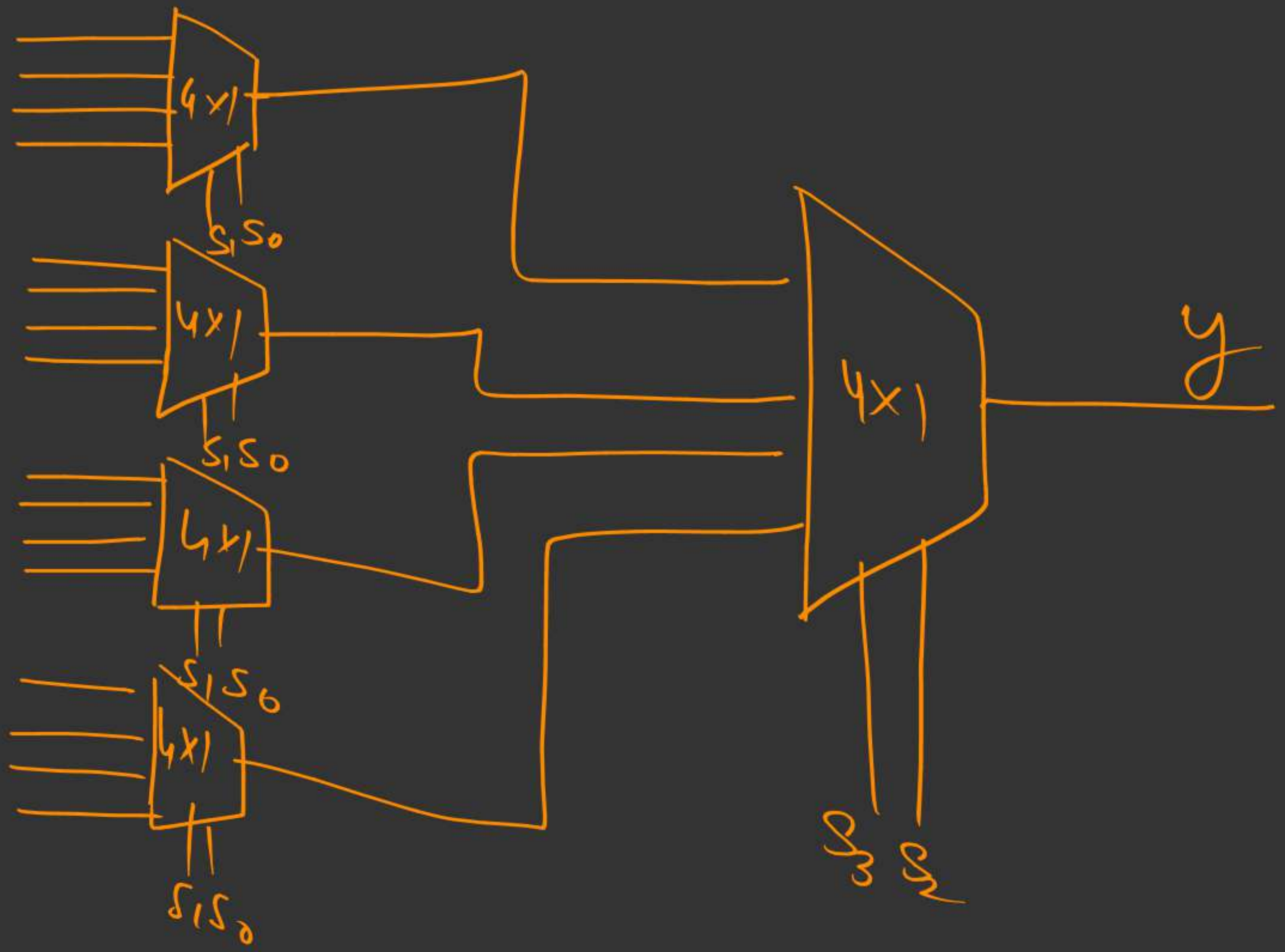
$16 \times 1 \longrightarrow 15$

$64 \times 1 \longrightarrow 63$

Ans : $\underline{\underline{(2^n - 1)}}$

(Q) How many 4×1 mux needed to design a 16×1 mux?

Logic



5

Shortcut : 16x1 using 4x1

$$\frac{16}{4} = 4$$

$$\frac{4}{4} = 1$$

Total = 4 + 1

= 5 (4x1 mux)

eg2:-

Design 64x1 ⟶ using 4x1 mux.

$16 \to 4 \to 1$

$\frac{64}{4} = 16$

$\frac{16}{4} = 4$

$\frac{4}{4} = 1$

Total $\Rightarrow 16+4+1$

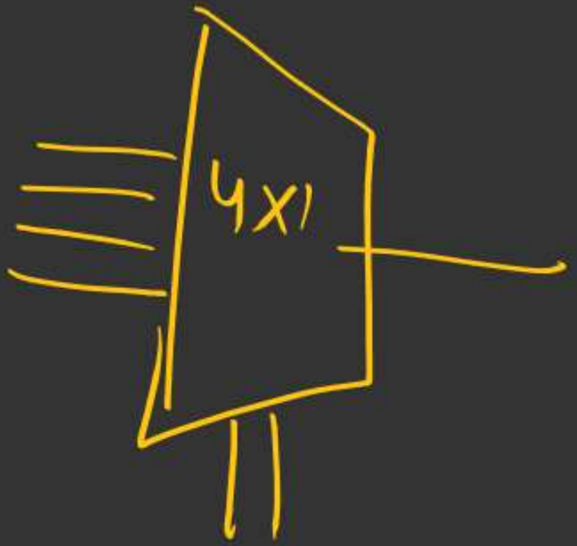
$= 21$

Special Cases (Shortcut still works)

eg3) Design 32x1 using 4x1 mux?

How many?

$32 \times 1 \longrightarrow$ using 4×1

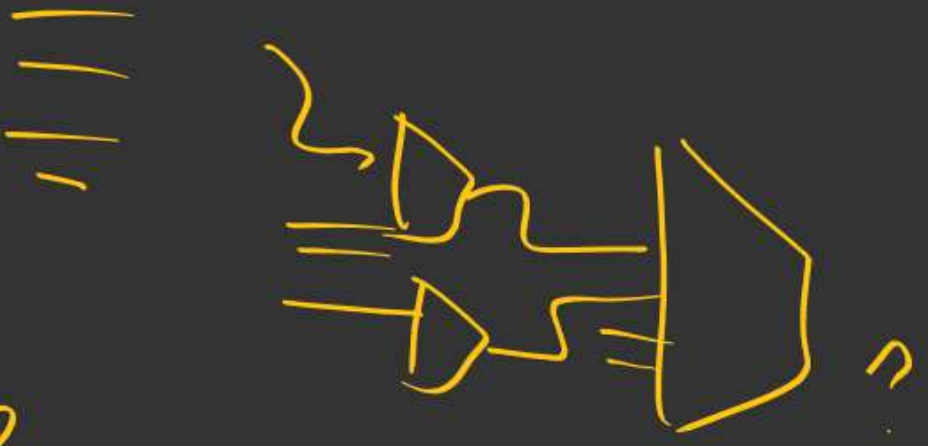


$$\frac{32}{4} = 8$$

$$\frac{8}{4} = 2$$

$$\frac{2}{4} \le 1$$ ✓ Stop

Total = $8 + 2 + 1 = 11$



Hw:

How many 8x1 mux needed for 64x1 ?

→ Think of internal working as well

2 mins Summary

→ MUX

- → Intro
- → Examples

THANK - YOU